A two-energy formula for the selected bilocal mean

Projection of the complete neutral tower

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Abstract

The single-selected first moment reads only a projection of the higher neutral fields. We compute this projection at every fixed degree using the established local source rules: the answer is an energy polynomial plus a one-U Wick module. Its recursion retains all lower Gaussian chaoses. The complete bilocal mean then reduces to the existing positive two-energy Mehler measure, with a factorial truncation bound for every fixed λ and finite time. This is a first-moment result; products of higher fields require their joint components outside the projected module.

1 Normalization and main result

Use the same overlapping complex Gaussian p-body CAR model and original disorder as the preceding stages [8, 5]. Take exact finite-N pinching first, the half-filled double scaling $p^2/N \to \lambda > 0$ next, and keep λ fixed here. Let $C = J_s^2/\lambda$, $q = e^{-4\lambda}$, $\zeta = \sqrt{q}$, $h = H/\sqrt{C}$, $W = \sqrt{\lambda}U$, where h, U are orthogonal unit-variance q-Gaussian fields. Throughout this note

$$t = \sqrt{C}\, t_{\mathrm{phys}}, \qquad T = \sqrt{C}\, T_{\mathrm{phys}}, \qquad \alpha_T(X) = e^{iTh} X e^{-iTh}, \qquad \hbar = 1.$$

These are h-times; the preceding string variable is $u = 2\sqrt{\lambda}\, T$. Write H_j for monic unit-variance q-Hermite polynomials and $p_j = [j]_q!$, so $\|H_j\|_2^2 = p_j$. The inherited single-selected density on the full marked core is

$$\rho = 1 - \zeta(1+\lambda) + \zeta(h-W)^2, \qquad \sigma(X) = \tau(\rho X). \tag{1}$$

It is the already identified positive normalized state, not an energy-only replacement. Put

$$d\nu(x,y) = K_q(x,y)\, d\mu_q(x)\, d\mu_q(y), \qquad K_q = \sum_{n\ge 0} \frac{q^n H_n(x) H_n(y)}{p_n}.$$

This is a probability measure on $[-2/\sqrt{1-q}, 2/\sqrt{1-q}]^2$. The polynomials $\Xi_j(x,y)$ below define

$$L_j(t;x,y) = \int_0^t (t-s)\, \mathrm{sinc}\!\left(\frac{(t-s)(x-y)}{2}\right) e^{-is(x+y)/2} c_j(s)\, ds, \tag{2}$$

$$c_j(s) = p_j^{-1} \int e^{isx} H_j(x)\, d\mu_q(x). \tag{3}$$

Here $\mathrm{sinc}\, z = \sin z/z$, continuously extended at zero.

Theorem 1. *For every fixed $\lambda > 0$, $t \ge 0$ and real T, the actual double-scaled bilocal in the inherited normalization satisfies*

$$\boxed{\sigma(\mathcal{B}(T,t)) = 2\zeta\lambda \int (x+y) e^{iT(x-y)} \sum_{j\ge 1} \Xi_j(x,y) L_j(t;x,y)\, d\nu(x,y).} \tag{4}$$

The sum converges in the indicated integrated sense uniformly on fixed compact t windows. The degree-tail bound in Section 5 is uniform in real T. No N-uniform or growing-time limit is claimed.

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2 One marked insertion: an all-degree projection

Work in the joint tracial L^2 completion furnished by the marked moments. Let P project onto the closed polynomial space of h, U. This definition does not require boundedness of every higher field. Use $C = 1$ representatives

$$Z_n = \text{coupling-Wick}(h^n), \quad S_v = (vp - L)Z_v, \quad T_{v,w} = S_v \diamond Z_w, \quad \mathcal{W}_{a,b} = \text{Wick}_q(h^a U h^b).$$

The whole $v + w$ block is coupling-Wick; its first v slots carry the local-size mark. $\diamond$ retains their original order. The size superoperator L is distinct from $L_j(t; x, y)$ in (2). Set

$$\kappa_v = \lambda(v^2 - 4q\partial_q \log[v]_q!).$$

Theorem 2. *For $v \geq 1$, $w \geq 0$, $n = v + w$,*

$$\mathsf{P}T_{v,w} = \kappa_v H_n(h) + \sqrt{\lambda} \sum_{i=0}^{v-1} \mathcal{W}_{i,n-1-i}. \tag{5}$$

Every centered pure charged field J_d has $\mathsf{P}J_d = 0$. These are projection identities, not identities of the original fields with the displayed polynomials.

Proof. Test against an arbitrary word in h, W and expand its joint moments by the established disjoint-mark source rule [4, 5]. All source cumulants beyond degree two vanish: a surviving source consumes at least two of the four incidences of a two-label collision. A centered single-slot W has zero singleton source. Two such slots have connected coefficient one exactly when they are the ends of the same Gaussian chord.

The connected coefficient between the L block and a W slot is one precisely when that slot's partner lies in the marked prefix. It is zero otherwise. Indeed the W source can survive only its own two-label collision. At its other end $L_i n = n - I/2$, giving the quadratic source coefficient one after singleton normalization. A whole-D block instead has coefficient zero, since $D_i n = n$. These statements use local CAR algebra, not an independent probe replacement.

There are now exactly two cases. If the block mark is a singleton in the source partition, all external W slots pair among themselves. Fix their chords and the external partners of the source slots. Permuting the first v source endpoints has Mahonian weight $[v]_q!$; crossings with outside spectator chords do not depend on this permutation. Averaging $\lambda(v^2 - 4\,\text{cr}_v)$ gives κ_v. This is exactly the known energy regression times the unmarked whole-Wick block. If the block mark pairs with one external W, all other W slots pair among themselves. Choose the incident source slot $i < v$: replacing it by a W gives exactly the same pairing, crossing factor and factor λ. Summing over i is the second term of (5), since $W = \sqrt{\lambda}U$.

The multiblock q-Wick rule [2] identifies these restricted pairing sums with the displayed Wick polynomials. For a charged block the second case is absent; its known energy regression is $\lambda[d(d-1) - 4q\partial_q \log[d]_q!]H_d$. Subtraction gives zero. Equality against every polynomial proves the L^2 projections. □

3 Propagating the projection through the neutral tower

Two compatibility points matter. First, for a polynomial test A, $\tau(A^\dagger hX) = \tau((hA)^\dagger X)$, and similarly on the right. Thus the projected finite-degree expressions respect left and right multiplication by h. This follows at the moment core; it does not posit a bounded multiplication map for all higher fields.

Second, the coupling-chaos grading is orthogonal already at finite N. For an ordinary field block its degree is the number of chords leaving it. The q-Wick cancellation of internal pairs identifies the d-slot colored Wick vector with degree d in the joint limiting Gram space. Pairings between two whole-Wick blocks of unequal lengths vanish, including marked blocks. Expanding any test into these colored Wick vectors therefore gives

$$\mathsf{P}\,\mathbf{1}_{\mathcal{N}=d}X = \mathbf{1}_{\mathcal{N}=d}\mathsf{P}X, \qquad \mathsf{P}q^{\mathcal{N}}X = q^{\mathcal{N}}\mathsf{P}X \tag{6}$$

for the finite core used here. One may also prove this directly by fixing all internal partial matchings before the proof of Theorem 2; each matching fixes the outgoing degree. In particular one must not apply $q^{\mathcal{N}}$ as an energy function.

The inherited local recursion [6] is

$$r_j = 4\lambda \sum_{v=1}^{j-1} q^v T_{v,j-1-v}, \qquad \eta_{j+1} = h\eta_j - [j]_q \eta_{j-1} - r_j, \tag{7}$$

$$V_j = 4\lambda q^j S_j + (q^j - q^{\mathcal{N}})\eta_j, \qquad N_j = V_j - v_j(h), \tag{8}$$

with $\eta_0 = \eta_1 = \eta_2 = 0$ and v_j the established energy regression. Write $\mathsf{P}\eta_j = g_j(h) + \sqrt{\lambda}E_j$ and define

$$E_0 = E_1 = E_2 = 0, \tag{9}$$

$$E_{j+1} = hE_j - [j]_q E_{j-1} - 4\lambda \sum_{v=1}^{j-1} q^v \sum_{i=0}^{v-1} \mathcal{W}_{i,j-2-i}, \tag{10}$$

$$\Xi_j = 4\lambda q^j \sum_{i=0}^{j-1} \mathcal{W}_{i,j-1-i} + (q^j - q^{\mathcal{N}})E_j. \tag{11}$$

Then (5)–(6) give

$$\mathsf{P}N_j = \sqrt{\lambda}\Xi_j, \qquad q^{\mathcal{N}}\mathcal{W}_{a,b} = q^{a+b+1}\mathcal{W}_{a,b}. \tag{12}$$

The scalar g_j and all its lower chaoses cancel only after the final energy regression. They were not discarded inside the recursion. Multiplication in this module is explicit:

$$h\mathcal{W}_{a,b} = \mathcal{W}_{a+1,b} + [a]_q \mathcal{W}_{a-1,b} + q^{a+1}[b]_q \mathcal{W}_{a,b-1}. \tag{13}$$

Terms with a negative index vanish. The right rule exchanges a, b. For example,

$$E_3 = -4\lambda qU, \qquad E_4 = -4\lambda q(1+q)(hU + Uh),$$

$$\Xi_1 = 4\lambda qU, \qquad \Xi_2 = 4\lambda q^2(hU + Uh),$$

$$\Xi_3 = 4\lambda q^3(\mathcal{W}_{2,0} + \mathcal{W}_{1,1} + \mathcal{W}_{0,2}) + 4\lambda q^2(1-q^2)U.$$

The last term is a genuine lower-chaos contribution to the projection.

4 Two-energy realization and the actual time filter

The one-U module is isometric to $L^2(\nu)$ by $h^a U h^b \mapsto x^a y^b$: the inner product is the two-U q-Wick Mehler identity. In this representation

$$\mathcal{W}_{a,b} = P_{a,b}(h_L, h_R)U, \quad P_{0,b} = H_b(y), \quad P_{a+1,b} = xP_{a,b} - [a]_q P_{a-1,b} - q^{a+1}[b]_q P_{a,b-1}.$$

Thus $P_{1,1} = xy - q$, $P_{2,0} = x^2 - 1$. These are the existing Casper–Kolb–Yakimov polynomials [1], with normalization

$$P_{a,b}(x,y) = (1-q)^{-(a+b)/2} H^{\mathrm{CKY}}_{a,b}\left(\frac{\sqrt{1-q}x}{2}, \frac{\sqrt{1-q}y}{2}; q, q\right). \tag{14}$$

Equation (14) follows by matching the recurrence and initial polynomials. Their other orthogonality measures are not being identified with ν. Here the measure follows from the tracial two-U inner product. Below $\Xi_j(x,y)$ means the polynomial obtained from (9) by this isometry.

The already controlled actual memory limit [8] is, in h units, $\widehat{\mathcal{F}}(s) = \mathcal{F}_{\mathrm{phys}}(s/\sqrt{C})/C$, with neutral part

$$\widehat{\mathcal{F}}_N(s) = 2e^{-ihs/2} \sum_{j\geq 1} c_j(s) N_j e^{-ihs/2}.$$

The charged part is a convergent sum of h-dressed centered J_d's. Its projection is zero by Theorem 2. The selected density (1) belongs to the h, U subspace, so it reads only $\mathsf{P}\widehat{\mathcal{F}}$. Define

$$\Psi_s(x,y) = \sum_{j\geq 1} c_j(s)\Xi_j(x,y).$$

The energy-only and $\lambda(U^2-1)$ terms in ρ have zero overlap with a one-U field. Its odd term $-\zeta\sqrt{\lambda}\{h,U\}$ gives exactly

$$\sigma(\alpha_T \widehat{\mathcal{F}}(s)) = -2\zeta\lambda \int (x+y) e^{iT(x-y) - is(x+y)/2} \Psi_s(x,y)\, d\nu. \tag{15}$$

Both $\sqrt{\lambda}$ factors are retained: one from the density, one from (12).

The actual Volterra reconstruction, including its sign, is

$$\mathcal{B}(T,t) = -\int_0^t ds \int_{-(t-s)/2}^{(t-s)/2} dv\, \alpha_{T+v}\widehat{\mathcal{F}}(s), \qquad t \geq 0. \tag{16}$$

The change from physical to h units cancels the factor C against the two time differentials. Integrating $e^{iv(x-y)}$ over the inner interval gives $(t-s)\,\mathrm{sinc}((t-s)(x-y)/2)$, proving (4) once the convergence below is supplied. Negative times follow by the inherited adjoint convention. No multiplication of projected higher fields is used in this first-moment argument.

5 A usable all-degree tail bound

Put $r = (1-q)^{-1/2}$, $F_q = 4\lambda q/(1-q)$ and $R_\rho = \sqrt{1+q(1+q)(1+\lambda)^2} = \|\rho\|_2$. Every one-$U$ Wick vector of length n has norm at most $\sqrt{[n]_q!} \le r^n$, by summing its color-compatible permutations. The source in the E_{j+1} recursion has norm at most

$$4\lambda r^{j-1} \sum_{v\ge 1} v q^v = \frac{4\lambda q}{(1-q)^2} r^{j-1}.$$

Multiplication by h in this module has norm at most $2r$. Consequently $z_j = \|E_j\|_2/r^j$ obeys $z_{j+1} \le 2z_j + z_{j-1} + F_q$, with $z_0 = z_1 = z_2 = 0$. Induction gives $z_j \le F_q 3^j$. The multiplier $q^j - q^{\mathcal{N}}$ has norm at most one, since its eigenvalues are differences of two numbers in $[0,1]$. Hence

$$\|\Xi_j\|_{L^2(\nu)} \le 4\lambda j (qr)^j + F_q (3r)^j. \tag{17}$$

Lemma 1. *For every real s, $|c_j(s)| \le |s|^j/j!$.*

Proof. Let $x_1, \dots, x_j$ be the real simple roots of H_j and let $I_{j-1}f$ interpolate $f(x) = e^{isx}$ there. The interpolation remainder is $H_j(x) f[x_1, \dots, x_j, x]$. Since H_j is orthogonal to $I_{j-1}f$,

$$c_j(s) = p_j^{-1} \int H_j(x)^2 f[x_1, \dots, x_j, x] d\mu_q(x).$$

The Hermite–Genocchi simplex formula [3] bounds the divided difference by $\sup_{\mathbb{R}} |f^{(j)}|/j! = |s|^j/j!$. The remaining positive weight has integral one. This also applies at coincident nodes by continuity. □

Combining the lemma with (17) proves absolute $L^2(\nu)$ summability on $|s| \le S$. For complex s the same argument adds only $e^{2r|\operatorname{Im} s|}$, giving locally uniform analytic series and justifying the short-time expansion below. Write $A = qrS$, $B = 3rS$ and define, for $J+1 > A$, $J+2 > B$,

$$\mathcal{R}_J(S) = \sqrt{\lambda} \left[\frac{4\lambda A^{J+1}}{J!\,[1 - A/(J+1)]} + \frac{F_q B^{J+1}}{(J+1)!\,[1 - B/(J+2)]} \right]. \tag{18}$$

This bounds the projected neutral sum after J, before its outer factor two. The time simplex has area $t^2/2$ and the phase multiplications are unitary on $L^2(\nu)$, so

$$|\sigma(\mathcal{B}(T,t)) - \sigma(\mathcal{B}_{\le J}(T,t))| \le t^2 R_\rho \mathcal{R}_J(S), \qquad 0 \le t \le S. \tag{19}$$

This is an analytic degree bound, not a floating-point error estimate.

To identify the series with the microscopic object we use the already proved actual/local tails and sector/pinching bridge [7, 6, 8], not just (17) for its limit. The inherited tail is $K_S \sum_{j>J} (j+1)^2 X_S^j/\sqrt{j!}$; it vanishes before removing the finite-N cutoff. The established preparation-density bounds transfer that convergence to σ. Thus the order is fixed degree, double scaling, then removal of degree cutoff. No uniform small-λ estimate is inferred.

6 A nonzero mean and executed checks

The first terms are

$$\Psi_s = 4i\lambda qs - 2\lambda q^2 s^2(x+y) + O_{L^2}(s^3).$$

The common outer phase in (15) must also be expanded. Its linear term changes the quadratic coefficient to $2\lambda q(1-q)s^2(x+y)$. Put

$$H_q(T) = \int (x+y)^2 e^{iT(x-y)}\, d\nu, \qquad H_q(0) = 2(1+q).$$

Symmetry makes the linear contribution zero. The original N_j are self-adjoint, so their projected symbols are symmetric in x, y; their parity is $(-1)^{j-1}$. Together with the parity of c_j these symmetries make the mean even in s and in T. The convergent series therefore gives, uniformly on compact T,

$$\sigma(\alpha_T \widehat{\mathcal{F}}(s)) = -4\zeta\lambda^2 q(1-q) H_q(T) s^2 + O(s^4), \tag{20}$$

$$\sigma(\mathcal{B}(T,t)) = \frac{\zeta\lambda^2 q(1-q)}{3} H_q(T) t^4 + O(t^6), \tag{21}$$

$$\boxed{\sigma(\mathcal{B}(0,t)) = \frac{2}{3}\sqrt{q}\,\lambda^2 q(1-q^2)t^4 + O(t^6).} \tag{22}$$

The coefficient is positive for physical $0 < q < 1$. An energy-only density would give zero for this complete first moment, since every surviving projection contains one U.

Exact checks. The local CAR source evaluator was compared with an independent colored q-Wick enumeration and the projected recursion. The first successful run saved 196 prefix identities, 40 charged identities, 44 chaos-filter identities, 32 original-descendant comparisons, and 8 full-selected boundary checks: 320 in total. Parameters $(q, \lambda) = (1/16, 1/3), (16/81, 1/5)$ are formal rational tests of algebra, not points on $q = e^{-4\lambda}$. Another driver verified 72 polynomial identities against the CKY explicit sum, and 10 force/commutator Taylor coefficients against the original fields with selected boundary sources. It checked the bilocal t^4 coefficient and retained the t^6 coefficients.

Finite-time diagnostics. For physical $q = 1/4, 1/2$ and $\lambda = -\log q/4$, 12 time pairs were evaluated with $(J, L, n_s) = (24, 48, 24)$ and $(32, 64, 32)$. Here L is the energy Jacobi size and n_s the time quadrature order. All differences were below $2.78 \cdot 10^{-17}$.

q	t	T	$\operatorname{Re}\sigma(\mathcal{B}(T,t))$
1/4	1/2	0	0.0005532382640754
1/4	1	0	0.0074594258627641
1/4	1	1/2	0.0066385092055227
1/2	1	0	0.0043015230840990

The largest degree bound (19) in these refined rows is $2.79 \cdot 10^{-17}$. It excludes spectral/time quadrature and roundoff. Cancellation factors reach about 37.7. The scalar c_j sums start at order j, avoiding cancellation of tiny high-degree coefficients in eigensolver matrix elements. Raw values, refinement differences, exact-arithmetic degree bounds, versions and input hashes are saved separately. These diagnostics do not certify total numerical errors.

7 Interpretation and continuation

This stage closes the full neutral-tower contribution to a single-selected bilocal *mean*. The charged part has zero projection for this purpose. It does not follow that charged fluctuations, charged–neutral covariances, or components orthogonal to h, U vanish. In particular $\mathsf{P}(XY) \neq \mathsf{P}(X)\mathsf{P}(Y)$ in general. Computing selected bilocal two- and four-point functions requires new multiple-mark projections or the full established joint source calculus.

The current result is a microscopic fixed-λ model calculation with the actual preparation and pinching interface. It is not yet a bulk mass/position dictionary, a finite-G scattering result, or a completed de Sitter dual. The next direct calculation is a two-insertion projection with the connected L–L and D–L source terms retained, followed by the same actual time filters. Its low-degree limit must reproduce the previous neutral-slope two-time response [9].

References

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